

Calculus II Survival Sheet

Volumes of Revolution

Disk Method

$$V = \pi \int_a^b R^2 dx$$

Washer Method

$$V = \pi \int_a^b (r_{outer}^2 - r_{inner}^2) dx$$

Shell Method

$$V = 2\pi \int_a^b (\text{radius})(\text{height}) dx$$

Rule:

- Parallel slices $\rightarrow$ Washers
- Perpendicular slices $\rightarrow$ Shells

Arc Length

$$L = \int_a^b \sqrt{1 + (y')^2} dx$$

Integration Techniques

Integration by Parts

$$\int u dv = uv - \int v du$$

Trigonometric Integrals

Odd power of sine: save one $\sin x$.

Odd power of cosine: save one $\cos x$.

Even powers: use identities

$$\sin^2 x = \frac{1 - \cos 2x}{2}, \quad \cos^2 x = \frac{1 + \cos 2x}{2}$$

Trig Substitution

$$\sqrt{a^2 - x^2} \Rightarrow x = a \sin \theta$$

$$\sqrt{a^2 + x^2} \Rightarrow x = a \tan \theta$$

$$\sqrt{x^2 - a^2} \Rightarrow x = a \sec \theta$$

Always draw a triangle to back-substitute.

Partial Fractions

Used to integrate rational functions:

$$\int \frac{P(x)}{Q(x)} dx$$

Requirement:

$$\deg(P) < \deg(Q)$$

If not, perform polynomial long division first.

Step 1: Factor the Denominator

Factor $Q(x)$ completely over the reals.

Case 1: Distinct Linear Factors

If

$$Q(x) = (x - a)(x - b)$$

then

$$\frac{P(x)}{(x - a)(x - b)} = \frac{A}{x - a} + \frac{B}{x - b}$$

Example:

$$\frac{3x + 5}{(x - 1)(x + 2)} = \frac{A}{x - 1} + \frac{B}{x + 2}$$

Case 2: Repeated Linear Factors

If

$$Q(x) = (x - a)^n$$

include every power:

$$\frac{P(x)}{(x - a)^n} = \frac{A_1}{x - a} + \frac{A_2}{(x - a)^2} + \cdots + \frac{A_n}{(x - a)^n}$$

Example:

$$\frac{2x + 1}{(x - 3)^3} = \frac{A}{x - 3} + \frac{B}{(x - 3)^2} + \frac{C}{(x - 3)^3}$$

Case 3: Irreducible Quadratic Factors

If the quadratic cannot factor over the reals:

$$x^2 + bx + c$$

use a linear numerator:

$$\frac{Ax + B}{x^2 + bx + c}$$

Example:

$$\frac{x}{(x-1)(x^2+4)} = \frac{A}{x-1} + \frac{Bx+C}{x^2+4}$$

Case 4: Repeated Irreducible Quadratics

If

$$(x^2 + 1)^2$$

then

$$\frac{Ax + B}{x^2 + 1} + \frac{Cx + D}{(x^2 + 1)^2}$$

Improper Integrals

Improper integrals are defined using limits.

Type I: Infinite Interval

Infinite upper bound

$$\int_a^\infty f(x) dx = \lim_{b \rightarrow \infty} \int_a^b f(x) dx$$

Infinite lower bound

$$\int_{-\infty}^a f(x) dx = \lim_{b \rightarrow -\infty} \int_b^a f(x) dx$$

Both directions

$$\int_{-\infty}^\infty f(x) dx = \int_{-\infty}^c f(x) dx + \int_c^\infty f(x) dx$$

Each limit must converge separately.

Type II: Infinite Integrand (Vertical Asymptote)

If $f(x)$ is discontinuous or undefined at an endpoint or inside the interval, rewrite using limits.

At right endpoint $x = b$:

$$\int_a^b f(x) dx = \lim_{t \rightarrow b^-} \int_a^t f(x) dx$$

At left endpoint $x = a$:

$$\int_a^b f(x) dx = \lim_{t \rightarrow a^+} \int_t^b f(x) dx$$

If discontinuous at $x = c$ inside (a, b) :

$$\int_a^b f(x) dx = \lim_{t \rightarrow c^-} \int_a^t f(x) dx + \lim_{t \rightarrow c^+} \int_t^b f(x) dx$$

Both limits must converge for the integral to converge.

Differential Equations (Separable + Initial Value Problems)

A differential equation relates a function and its derivatives.

We focus on **separable equations**:

$$\frac{dy}{dx} = g(x)h(y)$$

Step 1: Separate Variables

Rearrange to isolate x and y :

$$\frac{1}{h(y)} dy = g(x) dx$$

This is the key move: get all y terms with dy , all x terms with dx .

Step 2: Integrate Both Sides

$$\int \frac{1}{h(y)} dy = \int g(x) dx$$

Do NOT forget the constant:

$$+C$$

Usually written on one side after integrating (two constants become one new constant).

Step 3: Solve for y (if possible)

After integrating, isolate y explicitly if you can.

Sometimes you leave it implicit if solving is messy.

Initial Value Problems (IVPs)

Given:

$$\frac{dy}{dx} = g(x)h(y), \quad y(x_0) = y_0$$

Steps:

1. Separate variables
2. Integrate both sides
3. Apply initial condition (x_0, y_0)
4. Solve for constant C
5. Solve for y (if possible)

L'Hôpital's Rule

Used to evaluate limits of indeterminate forms:

$$\frac{0}{0} \quad \text{or} \quad \frac{\infty}{\infty}$$

If

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} \text{ is } \frac{0}{0} \text{ or } \frac{\infty}{\infty},$$

then:

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

Apply repeatedly if needed.

Important: Only use if the original limit is truly indeterminate.

Other Indeterminate Forms (must rewrite first)

L'Hôpital's Rule does NOT directly apply to:

$$0 \cdot \infty, \quad \infty - \infty, \quad 0^0, \quad 1^\infty, \quad \infty^0$$

These must be algebraically rewritten first.

Indeterminate Powers Strategy

For expressions of the form:

$$f(x)^{g(x)}$$

Take natural logs:

$$y = f(x)^{g(x)} \quad \Rightarrow \quad \ln y = g(x) \ln(f(x))$$

Then evaluate:

$$\lim \ln y = \lim g(x) \ln(f(x))$$

Now apply algebra + L'Hôpital if needed.

Finally exponentiate:

$$\lim y = e^{\lim \ln y}$$

Sequences

Definition 1.

A *sequence* is a function $a_n : \mathbb{N} \rightarrow \mathbb{R}$.

$$\{a_n\} = a_1, a_2, a_3, \dots$$

This is an infinitely long ordered list.

Limit of a Sequence

Definition 2.

The sequence $\{a_n\}$ has the *limit* L and we write

$$\lim_{n \rightarrow \infty} a_n = L.$$

If $\lim_{n \rightarrow \infty} a_n$ exists, we say the sequence *converges*. Otherwise, we say the sequence *diverges*.

Theorem 0.1. *If*

$$\lim_{n \rightarrow \infty} f(x) = L.$$

and $f(n) = a_n$ where n is an integer, then $\lim_{n \rightarrow \infty} a_n = L$.

Example: $1, -1, 1, -1, \dots$ is of the form $(-1)^{n+1}$

Example: $-1, 1, -1, 1, \dots$ is of the form $(-1)^n$

Algebraic Limit Laws

If $\lim a_n = A$ and $\lim b_n = B$, then:

$$\lim(a_n + b_n)$$

$$\lim(a_n b_n)$$

$$\lim \frac{a_n}{b_n}$$

Squeeze Theorem

If

$$a_n \leq b_n \leq c_n$$

and

$$\lim a_n = \lim c_n = L,$$

then

$$\lim b_n = L.$$

Infinite Series

Definition 3.

An infinite series is

$$\sum_{n=1}^{\infty} a_n.$$

Definition 4.

Define partial sums:

$$S_N = \sum_{n=1}^N a_n.$$

Example: These are partial sums:

$$s_1 = a_1$$

$$s_2 = a_1 + a_2$$

In general we have:

$$s_n = a_1 + a_2 + \dots + a_n$$

The series converges if $\lim_{N \rightarrow \infty} S_N$ exists.

Definition 5.

Given a series $\sum_{n=1}^{\infty} a_n = a_1 + a_2 + a_3 + \dots$, let s_n denote its n th partial sum.:

$$s_n = \sum_{i=1}^n a_i = a_1 + a_2 + a_3 + \dots + a_n.$$

If the sequence $\{s_n\}$ is convergent and $\lim_{n \rightarrow \infty} s_n = s$ exists as a real number, then the series $\sum_{n=1}^{\infty} a_n$ is called *convergent* and we write:

$$a_1 + a_2 + a_3 + \dots + a_n = s.$$

or

$$\sum_{i=1}^{\infty} a_i = s$$

The number s is called the *sum* of the series. If the sequence $\{s_n\}$ is divergent, the the series is called *divergent*.

Divergence Test

If

$$\lim_{n \rightarrow \infty} a_n \neq 0,$$

then the series diverges. If it is 0, no conclusion can be drawn.

Note: This test can be used on any series.

Linearity of Series

$$\sum_{n=1}^{\infty} c a_n = c \sum_{n=1}^{\infty} a_n, \quad c \in \mathbb{R}$$
$$\sum_{n=1}^{\infty} (a_n \pm b_n) = \sum_{n=1}^{\infty} a_n \pm \sum_{n=1}^{\infty} b_n$$

Telescoping Series

A telescoping series has cancellation between terms:

$$\sum (a_n - a_{n+1})$$

Steps:

- Rewrite using partial fractions if needed

- Expand partial sums
- Cancel terms
- Take limit of remaining boundary terms

Geometric Series

A *geometric series* is of the form:

$$\sum_{n=1}^{\infty} ar^{n-1} = a + ar + ar^2 + ar^3 + \dots + ar^{n-1} + \dots$$

where $a \neq 0$

Each term is obtained from the preceding one by multiplying it by the *common ratio* (r).

The geometric series

$$\sum_{n=1}^{\infty} ar^{n-1} = a + ar + ar^2 + ar^3 + \dots + ar^{n-1} + \dots$$

is convergent if $|r| < 1$ and its sum is:

$$\frac{a}{1-r}.$$

If $|r| \geq 1$ the geometric series is divergent.

Note: This is specifically for the geometric series.

p-Series

A *p-series* is of the form:

$$\sum_{n=1}^{\infty} \frac{1}{n^p}$$

- Converges if $p > 1$
- Diverges if $p \leq 1$

Harmonic Series:

$$\sum_{n=1}^{\infty} \frac{1}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots$$

Using the p-series test: $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges if $p > 1$, diverges if $p \leq 1$.
Here, $p = 1$, so the series diverges.

Integral Test

Let f be continuous from $[1, \infty)$, positive, decreasing and $a_n = f(n)$.

Then

$\sum a_n$ converges if and only if $\int_1^\infty f(x) dx$ converges

$\sum a_n$ diverges if and only if $\int_1^\infty f(x) dx$ diverges

either both converge or both diverge.

Comparison Tests

Direct Comparison Test

Suppose that $\sum a_n$ and $\sum b_n$ are series with positive terms.

- If $\sum b_n$ converges and $a_n \leq b_n \forall n \Rightarrow \sum a_n$ converges.
- If $\sum b_n$ diverges and $a_n \geq b_n \forall n \Rightarrow \sum a_n$ diverges.

Limit Comparison Test

Suppose that $\sum a_n$ and $\sum b_n$ are series with positive terms. If

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = c, \quad 0 < c < \infty,$$

then either both series converge or both diverge.

Alternating Series

Definition 6.

An *alternating series* is a series whose terms are alternately positive and negative.

Alternating Series Test

For

$$\sum (-1)^n b_n,$$

if:

- $b_{n+1} \leq b_n \forall n$,
- $\lim_{n \rightarrow \infty} b_n = 0$,

then the series converges.

Note: b_n is the absolute value of a_n

Absolute and Conditional Convergence

Definition 7.

A series is *absolutely convergent* if $\sum |a_n|$ converges.

Definition 8.

A series is *conditionally convergent* if it is convergent but not absolutely convergent ($\sum |a_n|$ diverges).

Theorem 0.2. *If a series $\sum a_n$ is absolutely convergent, then it is convergent.*

Ratio Test

Let

$$L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|.$$

- If $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = L < 1$ the series $\sum a_n$ is absolutely convergent (So it is convergent)
- If $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = L > 1$ the series $\sum a_n$ is divergent
- If $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = L = 1$ then the ratio test is inconclusive for the series $\sum a_n$.

Power Series

$$\sum_{n=0}^{\infty} c_n (x - a)^n$$

Radius of Convergence

Find R using Ratio Test.

- Converges for $|x - a| < R$
- Diverges for $|x - a| > R$
- Test endpoints separately

Power Series Operations

You can differentiate and integrate term-by-term:

$$\begin{aligned} \frac{d}{dx} \sum c_n (x - a)^n &= \sum n c_n (x - a)^{n-1} \\ \int \sum c_n (x - a)^n dx &= \sum \frac{c_n}{n+1} (x - a)^{n+1} \end{aligned}$$

Interval of Convergence Notes

Always express final answers as:

- $(a - R, a + R)$
- $[a - R, a + R]$
- mixed endpoints like $(a - R, a + R]$

Taylor Series

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x - a)^n$$

Maclaurin Series (Taylor Series with $a = 0$)

$$\begin{aligned} e^x &= \sum_{n=0}^{\infty} \frac{x^n}{n!}, \quad R = \infty \\ \sin x &= \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}, \quad R = \infty \\ \cos x &= \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}, \quad R = \infty \\ \frac{1}{1-x} &= \sum_{n=0}^{\infty} x^n, \quad |x| < 1, \quad R = 1 \\ \ln(1+x) &= \sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^n}{n}, \quad |x| < 1, \quad R = 1 \\ (1+x)^k &= \sum_{n=0}^{\infty} \binom{k}{n} x^n, \quad |x| < 1, \quad R = 1 \\ \tan^{-1} x &= \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1}, \quad |x| \leq 1, \quad R = 1 \end{aligned}$$

Parametric Functions

A parametric curve is defined by:

$$x = x(t), \quad y = y(t)$$

The parameter t traces the curve.

Slope of a Parametric Curve

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$$

Important:

$$\frac{dx}{dt} \neq 0$$

Tangent Line (Parametric)

At $t = t_0$:

$$x_0 = x(t_0)$$

$$y_0 = y(t_0)$$

Slope:

$$m = \left. \frac{dy/dt}{dx/dt} \right|_{t=t_0}$$

Equation:

$$y - y_0 = m(x - x_0)$$

Area (Parametric)

$$A = \int_{t=a}^{t=b} y(t) \frac{dx}{dt} dt$$

Arc Length (Parametric)

$$L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

Polar Functions and Regions

Polar form:

$$r = f(\theta)$$

Conversion:

$$x = r \cos \theta, \quad y = r \sin \theta$$

Slope in Polar Coordinates

$$\frac{dy}{dx} = \frac{\frac{dr}{d\theta} \sin \theta + r \cos \theta}{\frac{dr}{d\theta} \cos \theta - r \sin \theta}$$

Area in Polar Coordinates

$$A = \frac{1}{2} \int_{\alpha}^{\beta} r^2 d\theta$$

Between two curves:

$$A = \frac{1}{2} \int_{\alpha}^{\beta} (r_{\text{outer}}^2 - r_{\text{inner}}^2) d\theta$$

Arc Length (Polar)

$$L = \int_{\alpha}^{\beta} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta$$

Finding Area Between Polar Curves

Steps:

1. Find intersection angles ($r_1 = r_2$)
2. Determine outer and inner curve
3. use the formula